Automated Data Health & Profiling Dashboard

Remaining Work for Project Completion

Updated 2026-09-26

This list is derived from the explicit non-goals and final-form description in project_plan.md, updated to reflect what has been built since. It is not a confirmed roadmap - treat it as a starting point for planning the next phase(s) and confirm scope/priority with your reviewer.

Completed Since This List Was First Written

- Median imputation of missing numeric values before scoring (--no-impute restores the skip-incomplete-rows behaviour).
- Per-row “why flagged” explanations, and reporting of the model's actual score threshold.
- KNN as a second outlier detector (--method, --k), with standardised features and a side-by-side comparison of the two methods.
- Empirical check on a 200-row synthetic file with 8 planted outliers at contamination 0.05: KNN found 8/8, Isolation Forest 7/8 (both 8/8 at 0.10).

1. Explicitly Deferred in the Current Plan

project_plan.md lists these as non-goals for the CLI phase:

- No Streamlit UI - the plan states one already exists separately and "can be reconnected to this logic later."
- No preprocessing beyond median imputation - encoding and scaling remain explicitly out of scope.
- No report file export - all output is terminal-only by design so far.

2. Streamlit Front-End Integration

- Reconnect stats.py, ml_processing.py, and preprocessing.py to the existing (or a rebuilt) Streamlit app - these modules were written as pure functions returning data, specifically so they're reusable outside the CLI.
- Replace terminal printing with Streamlit widgets: st.dataframe for tables, st.metric for headline numbers (row/column counts, outlier count), charts for distributions.
- Replace the CLI's csv_path argument with a file-upload widget (st.file_uploader).
- Replace --method, --contamination, --k and --no-impute with interactive controls (selector, sliders, toggle).

3. Additional Preprocessing

- Categorical encoding (one-hot or label encoding) for non-numeric columns, if they need to feed into the ML steps.

- Feature scaling for the Isolation Forest input - KNN already standardises its own copy of the features, but Isolation Forest still splits on raw feature values, so columns on very different scales (e.g. income in the tens of thousands vs. tenure in single-digit years) can dominate its scoring.
- Additional imputation strategies beyond median (e.g. mean, most-frequent for categoricals, or model-based imputation) if median alone proves too crude for some datasets - the missed planted outlier in the evaluation is an example, since median-filling its blanked values hid what made it unusual.

4. Reporting & Export

- Export the full profile (overview, statistics, covariance matrix, flagged outliers, method comparison) to a file - PDF, HTML, or CSV - since the current phase is terminal-only by design.
- Separate export of just the flagged outlier rows for downstream review or ticketing.

5. Testing & Robustness

- Formal unit tests for stats.py, ml_processing.py, and preprocessing.py (current testing was manual, via sample CSVs run through the CLI, plus a one-off recall check against the planted outliers in sample_200.csv).
- Validation against larger, real-world datasets - current fixtures are small (5-200 rows) by design for quick inspection.
- Performance/memory handling for large CSVs if the tool needs to scale beyond small files (e.g. chunked reading); KNN's neighbour search is also costlier than Isolation Forest as row counts grow.

6. Possible Future Enhancements (Not Yet Scoped)

Not requested anywhere in the current plan - listed only as ideas worth raising with whoever owns the roadmap:

- Further outlier-detection methods beyond Isolation Forest and KNN (e.g. Local Outlier Factor, DBSCAN), or an ensemble that combines the methods' scores into a single ranking.
- Visual correlation heatmap in the Streamlit UI, complementing the numeric covariance matrix table.

7. Documentation

- A README covering setup, usage, and the CLI-to-Streamlit hand-off plan for whoever picks up the UI work.